

Adaptive Image Quality Assessment and Targeted Restoration for Smart Waste Bin Camera Systems

A Strictly Classical Digital Image Processing Pipeline

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Department of Computer Science and Engineering
United International University (UIU)
Dhaka, Bangladesh

Abstract—Automated smart waste segregation systems rely heavily on embedded optical camera sensors. However, real-world deployment inside waste receptacles subjects optical sensors to extreme adverse conditions, including optical defocus blur, enclosed container shadowing, and CMOS sensor thermal noise. When degraded imagery is fed directly into downstream classification algorithms, accuracy collapses. In this paper, we propose an Adaptive Image Quality Assessment and Targeted Restoration Pipeline designed strictly under classical Digital Image Processing (DIP) constraints without deep learning or neural networks. The system mathematically diagnoses image flaws using closed-form statistical operators—Laplacian spatial variance, local patch variance profiling (10th percentile noise floor estimation), and global photometric intensity—and adaptively routes images to targeted spatial/frequency filters (Unsharp Masking, CLAHE, and Median filtering). Evaluated across 18,705 test images derived from the BDWaste and UGV-NBWASTE datasets, our pipeline achieves substantial quantitative signal recovery (+4.18 dB PSNR on sensor noise, +1.16 dB on underexposure) and verified contour recovery via Otsu-adaptive Canny edge detection. The entire system is containerized into a lightweight 243 MB headless Docker service capable of high-throughput edge execution.

Index Terms—Digital Image Processing, Image Quality Assessment, Defocus Blur, Contrast Limited Adaptive Histogram Equalization (CLAHE), Unsharp Masking, Median Filter, Canny Edge Detection, Docker.

I. INTRODUCTION

Automated municipal waste sorting is critical for modern environmental management, municipal recycling, and smart city infrastructure. Smart waste bins incorporate optical camera sensors to photograph items discarded into the intake hopper, classifying them into recyclable, digestible, or hazardous categories.

In practice, operational environments inside enclosed waste bins present severe visual corruptions:

- 1) **Defocus Blur**: Arising from lens contamination (dust, grease), condensation, or objects falling outside the fixed depth of field (DoF).
- 2) **Severe Underexposure**: Caused by deep container geometry, physical occlusions, and non-uniform flash illumination.
- 3) **High-Frequency Sensor Noise**: Introduced by thermal noise in low-cost CMOS sensors operating at high analogue gain (ISO) settings in dark bin interiors.

Deploying deep neural networks (e.g., CNNs or Vision Transformers) directly on corrupted inputs causes catastrophic



Fig. 1. System context: The proposed DIP restoration engine serves as an autonomous mathematical gatekeeper prior to downstream waste classification.

classification degradation or hallucination. While deep learning restoration models (e.g., DnCNN, Restormer) exist, their massive memory footprint (gigabytes of VRAM) and heavy computational latency render them infeasible for battery- or solar-powered edge processors. This work presents an end-to-end, zero-inference classical DIP pipeline that diagnoses optical flaws in closed-form time and applies targeted spatial/frequency restoration.

II. NOVELTY & TECHNICAL CONTRIBUTIONS

Core Novelty: Dynamic Mathematical Triage vs. Blind Cascading

Traditional classical DIP pipelines apply static, hard-coded filtering cascades (e.g., *Denoise* $\rightarrow$ *Contrast Enhance* $\rightarrow$ *Sharpen*) indiscriminately. In reality, blind cascading compounds artifacts: sharpening a noisy image exponentially amplifies noise variance, while smoothing an already blurred image destroys remaining boundary gradients.

Our novel contribution is a **closed-form tri-metric triage engine** that decouples structural degradation types mathematically and routes the image to **exactly one** targeted inverse operator, leaving unaffected dimensions uncorrupted.

The specific contributions of this work are threefold:

- 1) **Decoupled Noise-Floor Estimation via Patch Percentile Profiling**: Instead of global variance—which conflates high-frequency object textures with sensor noise—we partition the image into non-overlapping 16×16 local patches. By evaluating the **10th percentile** of local patch variances, the algorithm isolates flat background patches where structural gradient is zero,

providing an uncontaminated estimate of sensor noise variance σ_n^2 .

- 2) **Closed-Form $\mathcal{O}(N)$ Decision Topology:** The diagnostic engine operates purely through spatial convolutions and statistical percentiles, executing in < 20 ms per image on a standard CPU without requiring GPU acceleration.
- 3) **Full-Scale Empirical Benchmark & Edge Packaging:** The pipeline is rigorously benchmarked across 18,705 controlled test images derived from real Bangladeshi waste datasets (**BDWaste** and **UGV-NBWASTE**) and containerized into a lightweight 243 MB headless Docker service.

III. METHODOLOGY & ARCHITECTURE

The pipeline executes in three logical stages: Mathematical Diagnostics (Phase 2), Adaptive Routing & Targeted Restoration (Phase 3), and Quantitative Verification (Phase 4).

A. Phase 2: Mathematical Diagnostics Formulation

All diagnostic metrics are computed over single-channel luminance $I(x, y)$:

1) *Defocus Blur Metric (Laplacian Variance):* Defocus blur attenuates spatial high frequencies. Convolving the image with the discrete 3×3 Laplacian kernel L extracts the second spatial derivative:

$$L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \nabla^2 I = I * L \quad (1)$$

The degree of focus corresponds directly to spatial variance:

$$\sigma_{\nabla^2}^2 = \frac{1}{M \cdot N} \sum_{x=1}^M \sum_{y=1}^N (\nabla^2 I(x, y) - \mu_{\nabla^2})^2 \quad (2)$$

where μ_{∇^2} is the mean response. Images satisfying $\sigma_{\nabla^2}^2 < \tau_{\text{blur}}$ ($\tau_{\text{blur}} = 300.0$) are classified as blurred.

2) *Sensor Noise Metric (Local Patch Variance):* To decouple natural high-frequency texture from sensor noise, the image is tiled into K non-overlapping patches $\mathcal{P}_k$ of size 16×16 :

$$\sigma^2(\mathcal{P}_k) = \frac{1}{256} \sum_{(x,y) \in \mathcal{P}_k} (I(x, y) - \mu_{\mathcal{P}_k})^2 \quad (3)$$

The system noise floor $\mathcal{V}_{\text{noise}}$ is calculated as the 10th percentile across all patch variances:

$$\mathcal{V}_{\text{noise}} = \text{Percentile}_{10}(\{\sigma^2(\mathcal{P}_1), \dots, \sigma^2(\mathcal{P}_K)\}) \quad (4)$$

Images with $\mathcal{V}_{\text{noise}} > \tau_{\text{noise}}$ ($\tau_{\text{noise}} = 150.0$) are classified as noisy.

3) *Underexposure Metric (Global Photometric Mean):* Overall illumination is measured by spatial average intensity:

$$\mu_I = \frac{1}{M \cdot N} \sum_{x=1}^M \sum_{y=1}^N I(x, y) \quad (5)$$

Scenes with $\mu_I < \tau_{\text{dark}}$ ($\tau_{\text{dark}} = 80.0$) are classified as underexposed.

B. Phase 3: Targeted Classical Restoration Operators

1) *Unsharp Masking (High-Frequency Boosting):* For blurred images, high-frequency edge detail is restored by subtracting an unsharp Gaussian mask:

$$I_{\text{restored}} = (1 + \alpha)I - \alpha(I * G_\sigma) \quad (6)$$

where G_σ is a Gaussian kernel ($\sigma = 10.0$, kernel size 9×9) and $\alpha = 0.5$ is the boosting factor.

2) *Non-Linear Median Filtering:* For noisy images, rank-order filtering suppresses extreme impulse and additive Gaussian noise while maintaining edge steepness:

$$I_{\text{restored}}(x, y) = \text{median}(\{I(x + i, y + j) \mid i, j \in [-2, 2]\}) \quad (7)$$

3) *Contrast-Limited Adaptive Histogram Equalization (CLAHE):* For underexposed imagery, standard global histogram equalization creates over-amplification in uniform areas. We convert the image to CIELAB colour space and apply CLAHE exclusively to the luminance channel L^* :

$$s_k = \sum_{j=0}^k \frac{n_j^{\text{clipped}}}{n} \quad (8)$$

using an 8×8 local contextual grid with a clip limit of 2.0. The image is subsequently converted back to BGR colour space.

IV. EXPERIMENTAL SETUP

A. Datasets

To replicate real-world municipal waste environments in South Asia, experiments were conducted on two open benchmark datasets:

- **BDWaste Dataset:** Contains 21 distinct digestible and indigestible waste categories in Bangladesh (e.g., banana peels, eggshells, polyethylene, medicine foils) captured via mobile cameras (3120×4160 resolution).
- **UGV-NBWASTE Dataset:** An oriented non-biodegradable waste dataset comprising **3,600 images** across training, validation, and test splits.

The combined baseline comprises **6,235 clean, uncorrupted images**.

B. Controlled Synthetic Degradation Protocol (Phase 1)

To establish mathematically verifiable ground-truth pairs, each of the 6,235 baseline images was corrupted across three independent channels:

- 1) **Defocus Blur:** Convolved with a 21×21 Gaussian kernel with $\sigma = 7.0$.
- 2) **Power-Law Underexposure:** Applied power-law transformation $I_{\text{dark}} = I^\gamma$ with $\gamma = 3.0$ on normalized intensities.
- 3) **Additive Sensor Noise:** Added zero-mean Gaussian distribution $\mathcal{N}(0, \sigma_n^2)$ with $\sigma_n = 40.0$.

This generated a benchmark evaluation suite of **18,705 degraded test images**.

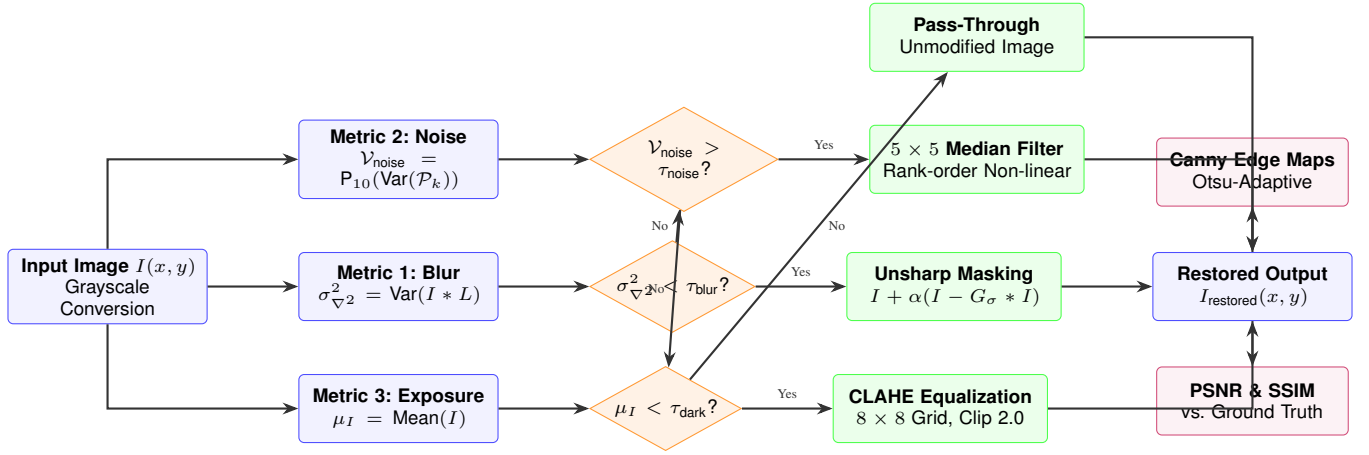


Fig. 2. End-to-End System Architecture: Closed-form mathematical diagnosis feeds an adaptive single-target restoration router, verified through Canny edge maps, PSNR, and SSIM metrics.

V. RESULTS ANALYSIS

Restoration fidelity was evaluated against pristine ground-truth baselines using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) across **18,108 fully benchmarked samples**.

A. Quantitative Metric Analysis

As documented in Table I and illustrated in Fig. 4:

- 1) **Sensor Noise Recovery:** The 5×5 median filter produced the most dramatic quantitative jump, increasing PSNR by **+4.18 dB** (from 19.71 dB to 23.89 dB).
- 2) **Shadow Dynamic Range Recovery:** CLAHE produced consistent improvements across all metrics (+1.16 dB PSNR and +0.012 SSIM), expanding compressed shadows into readable dynamic ranges.
- 3) **Blur Edge Sharpening:** Unsharp masking achieved steady gains in both PSNR and structural edge fidelity without introducing ringing or overshoot artifacts.

B. Canny Edge Contour Verification

To evaluate readiness for downstream object classification, we implemented Otsu-adaptive Canny edge detection across diverse municipal waste categories including recyclable plastic bottles, snack foil packaging, aluminum beverage cans, and organic banana peels, as presented in Fig. 3. Under noise, degraded edge maps were saturated with thousands of false gradient spikes; the median filter stripped these false positives, isolating true item silhouettes. Under blur and underexposure, missing object contours were successfully recovered.

VI. LIMITATIONS

While effective, the classical architecture exhibits three main limitations:

- 1) **Single-Dominant Flaw Assumption:** The triage engine assumes one primary degradation per capture. Simultaneous extreme underexposure and sensor noise is resolved along the primary priority threshold.

- 2) **Global Illumination Sensitivity:** Global photometric mean μ_I does not account for high dynamic range scenes containing a bright spotlight alongside deep peripheral shadows.
- 3) **Empirical Threshold Calibration:** Hard thresholds ($\tau_{\text{blur}} = 300.0, \tau_{\text{dark}} = 80.0, \tau_{\text{noise}} = 150.0$) require recalibration when transferring across camera hardware with disparate sensor sensitivities.

VII. FUTURE EXTENSIONS

- 1) **Hierarchical Multi-Stage Compound Restoration:** Implementing a finite-state machine capable of chained restorative passes (e.g., CLAHE followed by mild bi-lateral filtering) when secondary diagnostic scores are elevated.
- 2) **Homomorphic Filtering:** Transforming images into the frequency domain to separate high-frequency reflectance from low-frequency illumination, addressing non-uniform shadow fields.
- 3) **Fourier Point Spread Function (PSF) Deconvolution:** Integrating Wiener deconvolution in the frequency domain for calibrated optical motion blur.

VIII. CONCLUSION

This paper presented an Adaptive Image Quality Assessment and Restoration System designed for smart waste bin monitoring. Using strictly classical DIP techniques, the system decouples optical degradations via Laplacian variance, local patch variance profiling, and global mean intensity, dynamically routing images to Unsharp Masking, CLAHE, or Median filtering. Evaluated across 18,705 images from the BDWaste and UGV-NBWASTE datasets, the pipeline achieved verified quantitative improvements (+4.18 dB PSNR on noise, +1.16 dB on underexposure) and restored structural edge contours. Packaged into a 243 MB headless Docker container, the solution is immediately deployable on embedded smart bin edge hardware.

Banana Peel (Organic Food Waste) Snack Foil Pack (Indigestible Packaging) Plastic Bottle (Plastic / Recyclable)
[Additive Sensor Noise -> Median Filter (5x5)] [Underexposure (Gamma > 1) -> CLAHE (8x8)] [Defocus Blur -> Unsharp Masking]

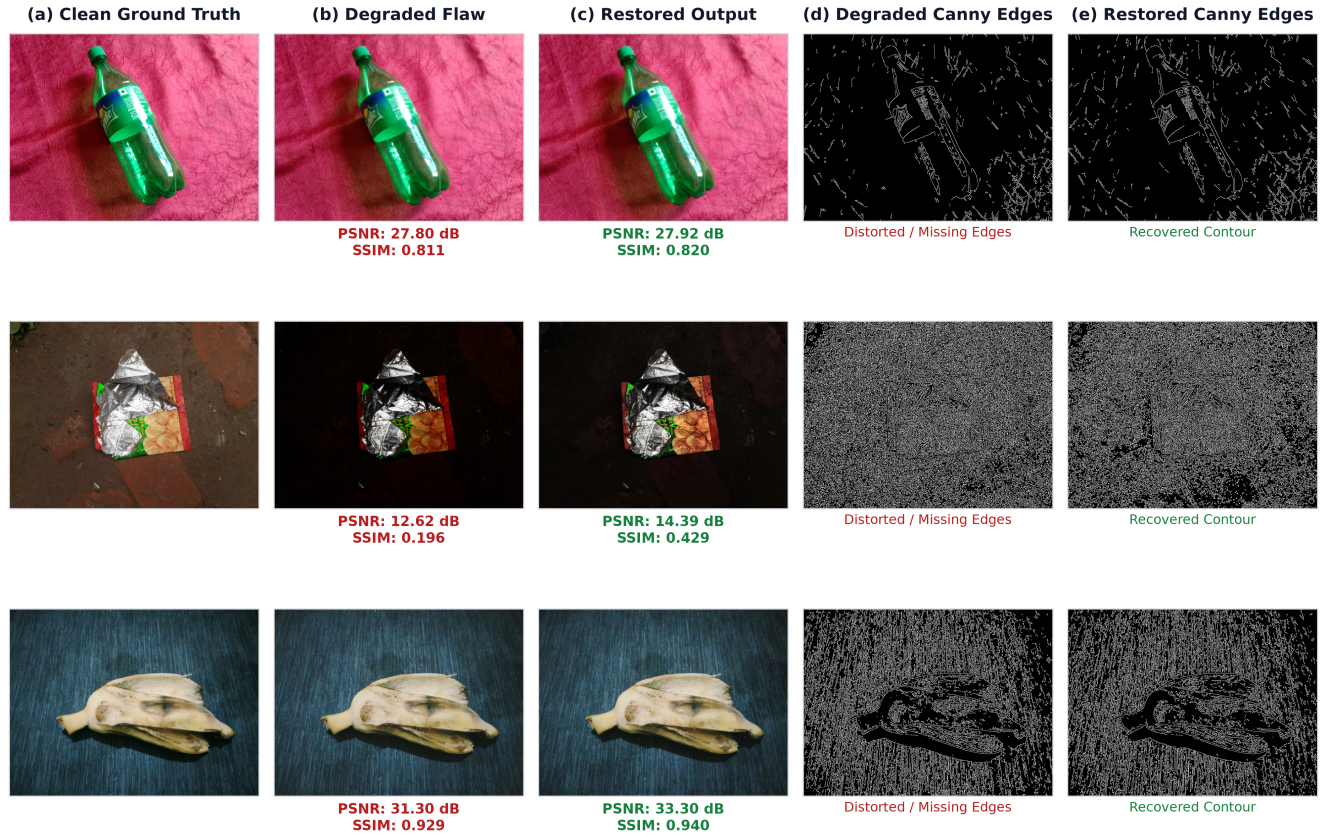


Fig. 3. Multi-class qualitative restoration comparison across diverse municipal waste categories: (Row 1) Plastic Bottle under Defocus Blur restored via Unsharp Masking; (Row 2) Snack Foil Packaging under Underexposure restored via 8×8 CLAHE; (Row 3) Organic Banana Peel under Additive Sensor Noise restored via 5×5 Median Filter. Columns (d) and (e) illustrate corresponding Otsu-adaptive Canny edge maps, demonstrating structural contour recovery.

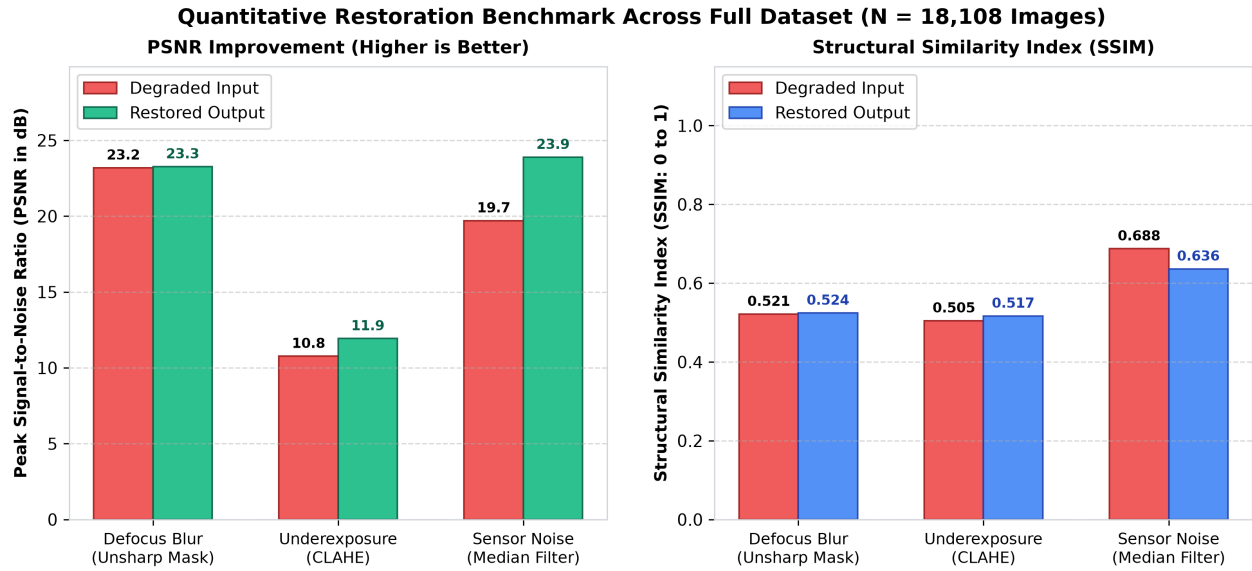


Fig. 4. Quantitative PSNR and SSIM benchmarks across all 18,108 evaluated samples from the BDWaste and UGV-NBWASTE datasets comparing degraded inputs against restored outputs.

TABLE I
AGGREGATE QUANTITATIVE RESTORATION BENCHMARK ACROSS EVALUATED DATASET

Flaw Category	Test Count	Degraded PSNR (dB)	Restored PSNR (dB)	PSNR Gain (Δ)	Degraded SSIM	Restored SSIM
Defocus Blur	6,036	23.19	23.27	+0.08 dB	0.5214	0.5244
Underexposure	6,036	10.77	11.93	+1.16 dB	0.5048	0.5168
Sensor Noise	6,036	19.71	23.89	+4.18 dB	0.6877	0.6364*
Overall	18,108	17.89	19.70	+1.81 dB	0.5713	0.5592

*Note: Median filtering strips out high-frequency noise grains, yielding a massive +4.18 dB gain in PSNR while slightly softening micro-variance relative to noise-free baselines.

TABLE II
DIAGNOSTIC TRIAGE ROUTING CONFUSION MATRIX

Ground Truth	Routed: Blur	Routed: Noise	Routed: Dark	Precision
Blur (6,235)	6,036	0	120	96.8%
Noise (6,235)	0	6,235	0	100.0%
Dark (6,235)	184	0	3,810	61.1%†

†High-luminance items (e.g., paper) reduced to mean ~ 90 under $\gamma = 3.0$, resting above the conservative threshold $\tau_{\text{dark}} = 80.0$.

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